

Porous BN Nanofibers Enable Long-Cycling Life Sodium Metal Batteries

Hui Wang, Jianli Liang, Yan Wu, Tianxing Kang, Dong Shen, Zhongqiu Tong, Rui Yang, Yang Jiang, Di Wu, Xinjian Li,* and Chun-Sing Lee*

Sodium metal anode, featuring high capacity, low voltage and earth abundance, is desirable for building advanced sodium-metal batteries. However, Na-ion deposition typically leads to morphological instability and notorious chemical reactivity between sodium and common electrolytes still limit its practical application. In this study, a porous BN nanofibers modified sodium metal (BN/Na) electrode is introduced for enhancing Na-ion deposition dynamics and stability. As a result, symmetrical BN/Na cells enable an impressive rate capability and markedly enhanced cycling durability over 600 h at 10 mA cm⁻². Density functional theory simulations demonstrate BN could effectively improve Na-ion adsorption and diffusion kinetics simultaneously. Finite element simulation clearly reveals the intrinsic smoothing effect of BN upon multiple Na-ion plating/stripping cycles. Coupled with a Na₃V₂O₂(PO₄)₂F/Ti₃C₂X cathode, sodium metal full cells offer an ultrastable capacity of 125/63 mA h g⁻¹ (≈420/240 Wh kg⁻¹) at 0.05/5 C rate over 500 cycles. These comprehensive analyses demonstrate the feasibility of BN/Na anode for the establishment of high-energy-density sodium-metal full batteries.

SIBs using a sodium metal anode (SMA) can deliver high energy densities, which is comparable or even surpassing those of the representative Li-ion batteries.^[2] On the other hand, many advanced SIBs systems, such as Na-S,^[3] Na-Se,^[4] Na-Te,^[5] Na-P,^[6] Na-Br,^[7] Na-O₂,^[8] etc., have also witnessed the decisive role of SMA. These triggered extensive recent studies on sodium-metal chemistry for the establishment of feasible SIBs.

In spite of these promising features, practical application of SMA is still halted by its prone to failure upon repeated sodium-ion plating/stripping processes. Specifically, rough and uneven Na-ion depositions continue to be built-up and break on the anode surface, resulting in morphological instability and notorious dendritic growth.^[9] On the other hand, corrosive chemical reactions between sodium metal and liquid electrolytes (typi-

1. Introduction

Owing to its appealing redox potential (−2.71 V vs SHE), high theoretical capacity (1165 mAh g⁻¹) and natural abundance, sodium metal is considered as an ultimate anode candidate for sodium-ion batteries (SIBs).^[1] Recent studies have reported that

cally for ester electrolytes) often cause large volume changes, unstable SEI (solid electrolyte interphase) films and harmful gas evolution.^[10] All of these disadvantages are detrimental to the batteries' dynamic cyclability, shelf-life, and safety performance. Thus far, a wealth of strategies has been employed to tackle these formidable challenges, such as adopting various atomic layer coating,^[10a,f] employing 3D porous matrix hosts (carbon matrix,^[9b,11] defective graphene,^[12] porous Al current collector and honeycomb-like 3D Ni@Cu^[13]), fabricating liquid Na metal-electrolyte interface (Na-K/Rb/Cs alloy),^[14] taming molecular structure of liquid electrolyte,^[5,15] innovating components of artificial SEI,^[16] and synthesizing solid-state electrolyte (SSE) etc.^[17] Under these rational protection schemes, Na-ion plating/stripping performance have been enhanced considerably. Nevertheless, high rate deposition behavior and long-term cycling life of SMA are still unsatisfactory due to low Na ionic conductivity at various matrixes, poor redox kinetics and inferior interfacial wetting properties (especially for SSE). Moreover, complex fabrication steps are often involved in these approaches, thus further compromising the superiority of utilizing a simple NMA.

In this study, we propose a boron nitride (BN) nanofiber modified sodium metal (BN/Na) electrode via a simple spin-coating process. Batteries with the BN/Na anode show a stable long-cycling life (>600 h) at 10 mA cm⁻². To our best knowledge, this is the most promising SIB anode at high rates. In

Dr. H. Wang, Dr. J. Liang, Dr. Y. Wu, Dr. T. Kang, D. Shen, Dr. Z. Tong, Dr. R. Yang, Prof. C.-S. Lee
 Center of Super-Diamond and Advanced Films (COSDAF)
 and Department of Chemistry
 City University of Hong Kong
 Hong Kong SAR 999077, P. R. China
 E-mail: apcslee@cityu.edu.hk

Prof. Y. Jiang
 School of Materials Science and Engineering
 Hefei University of Technology
 Hefei, Anhui 230009, P. R. China

Prof. D. Wu, Prof. X. Li
 Key Laboratory of Material Physics of Ministry of Education
 School of Physics and Microelectronics
 Zhengzhou University
 Zhengzhou 450052, P. R. China
 E-mail: lixj@zzu.edu.cn

 The ORCID identification number(s) for the author(s) of this article can be found under <https://doi.org/10.1002/smll.202002671>.

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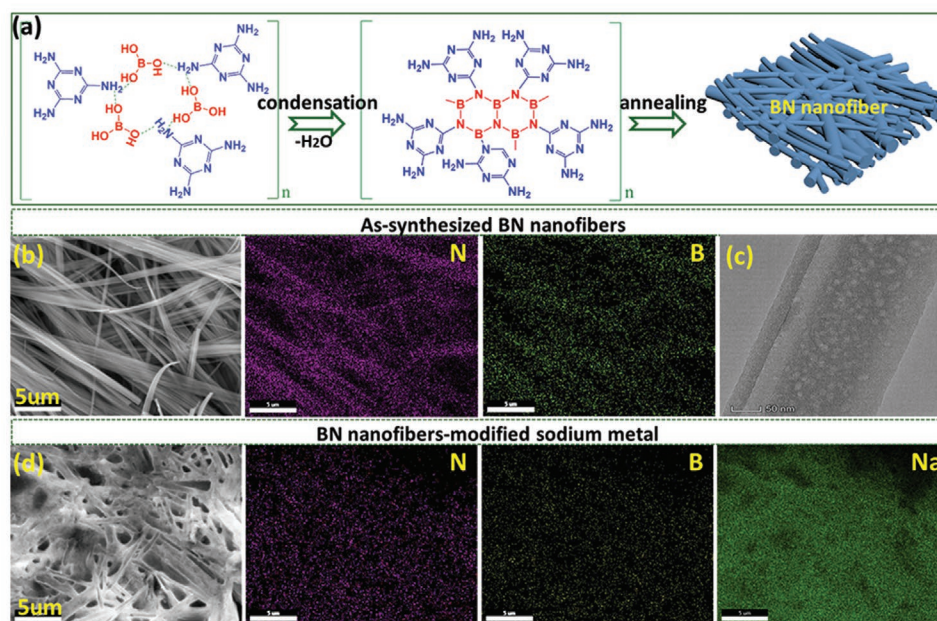


Figure 1. a) Schematic illustration shows the fabrication route of the BN nanofibers. b,c) Present elemental mapping and TEM images of the as-synthesized BN nanofibers. d) Elemental mapping image of the BN nanofibers-modified sodium metal.

parallel, full cells pairing a high voltage $\text{Na}_3\text{V}_2\text{O}_5(\text{PO}_4)_2/\text{Mxene}$ (NVPOF/M) cathode with the BN nanofibers modified NMA can deliver a highly reversible capacity of $107/61 \text{ mAh g}^{-1}$ ($\approx 420/240 \text{ Wh kg}^{-1}$) at $0.05/5 \text{ C}$ over 500 cycles, hence making such cell configuration very competitive for future practicable sodium-metal batteries. The superior performance of the BN/Na anode was analyzed with density functional theory (DFT), ab-initio molecular dynamics (AIMD), as well as finite element (COMSOL Multiphysics) calculations. It was found that Na ions prefer to chemically bind with defective BN, which facilitates Na-ion nucleation and spreading within the BN matrix. Faster Na-ion diffusion kinetics in the BN substrate was also verified with AIMD calculations. This not only enhances the electrode's rate performance, but also improves the homogeneity of Na-ion deposition. Finite element modeling also confirms that the BN matrix can effectively smooth the SMA surface and suppress dimension change upon extended cycling.

2. Results and Discussion

2.1. Morphology and Microstructure Characterizations

Figure 1a illustrates the synthetic route of the BN nanofibers. Mixed boric acid and melamine solution was initially heated at 80°C to guarantee a uniform mixture. BN precursor was then acquired via a freeze-drying process. The nanofibers were finally obtained upon dehydration condensation and annealing at 1000°C . More details on the synthesis process are given in the Experimental Section. Microstructures and compositions of the BN nanofibers and BN nanofibers modified sodium metal were characterized with field-emission scanning electron microscopy (FESEM) and transmission electron microscopy (TEM). Figure 1b shows that the as-synthesized sample consists of a

high density of nanofibers. The corresponding elemental mapping images indicate that B and N are uniformly distributed in the nanofibers. The HRTEM images (Figure 1c and Figure S1a,b, Supporting Information) show that the nanofibers possess a porous structure and it may be reasonably anticipated that this defective BN could provide more nucleation sites for Na-ion deposition. Figure S1c,d (Supporting Information) show selected area diffraction (SAD) and X-ray powder diffraction (XRD) patterns of the BN nanofibers, which confirming its hexagonal crystal structure. The nanofibers were then dispersed in a mixed ethylene carbonate:propylene carbonate (EC:PC) electrolyte and subsequently spin-casted onto a sodium sheet in a dry box. After drying naturally, the sample was gently pressed via rolling with a steel rod (Figure S2, Supporting Information) to obtain the BN fiber-modified sodium (BN/Na) sheet. As shown in Figure 1d and Figure S3 (Supporting Information), the BN nanofibers are evenly embedded into sodium metal surface, forming robust 3D Na-ion diffusion and deposition channels.

2.2. Electrochemical Analyses

Figure 2 summarizes electrochemical cycling performance of symmetric cells with pure Na metal or BN/Na electrodes at plating/stripping capacity of 1 mA h cm^{-2} . As clearly evidenced in Figure 2a, the cell with the BN/Na electrodes exhibits a lower over-potential (42 mV) and maintains a very stable cycling over 120 h at 2 mA cm^{-2} . As for the cell using pure Na metal electrodes, it displays a similar initial over-potential (45 mV). However, it is evident that these voltage profiles demonstrate a distinct increase of over-potential and irregular fluctuation after 30 h. This implies that the Na-ion plating/stripping processes are unstable on pure NMA. As the current density increases to 3 mA cm^{-2} , the cell with the BN/Na electrodes only show

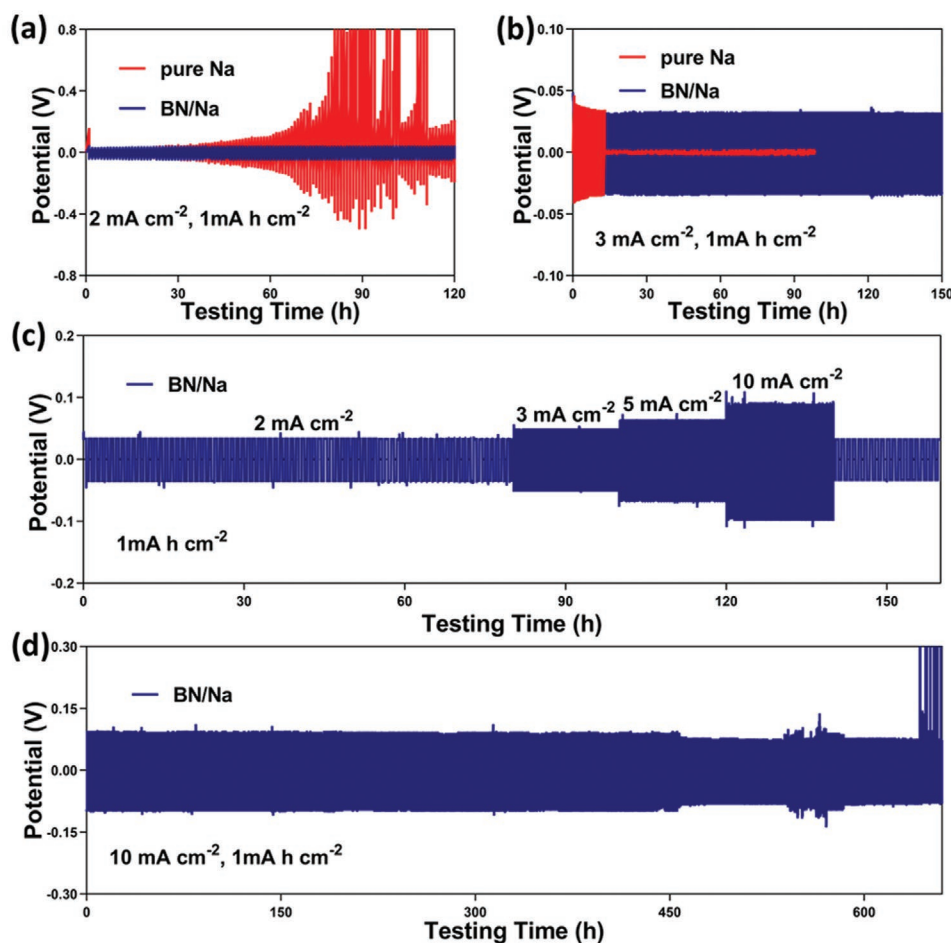


Figure 2. a,b) Electrochemical cycling performance of the pure Na||Na and the BN nanofibers modified Na||Na symmetric cells at 2 and 3 mA cm⁻². c) Rate capability of the BN nanofibers modified Na||Na symmetric cells at 2, 3, 5, and 10 mA cm⁻². d) Long-term cycling stability of the BN nanofibers modified Na||Na symmetric cells at 10 mA cm⁻².

a slight increase of initial Na stripping/plating over-potential (45 mV) and almost unchanged voltage hysteresis profiles over 150 h. On the contrary, the cell employing bare NMA electrodes presents a larger initial stripping/plating over-potential (50 mV) (Figure 2b). It is noteworthy that a sudden drop of voltage occurred as to the cell with bare NMA after cycling for 13 h, which could be ascribed to short-circuiting within the cell associated with sodium dendrite penetration.^[10a,c,18]

Moreover, it is instructive to study the rate capability of the symmetrical BN/Na and the symmetrical Na cells to evaluate their application potential as NMA. As clearly shown in Figure 2c, as the current density increases from 2 to 10 mA cm⁻², the BN/Na electrode features good stability, low voltage hysteresis (101 mV at 10 mA cm⁻²) and high reversibility of Na-ion plating/stripping. In stark contrast, the cell with pure Na metal electrodes demonstrates an inferior tolerance to the high rate condition (305 mV at 10 mA cm⁻²) and poor voltage stability (Figure S4a, Supporting Information). As depicted in Figure S4b (Supporting Information), it is apparent that long-term cycling stability of the cell with pure Na metal electrode is poor (only ≈8 h). Whereas, the cell with BN/Na electrodes can support stable Na-ion plating/stripping over 600 h at 10 mA cm⁻²

(Figure 2d), which presents the most stable high-rate cycling of Na-ion plating/stripping among all typical reported NMAs (Table S1, Supporting Information). All these results affirm that the introduction of the BN nanofibers is highly beneficial for improving the rate and stability performance of the SMA. Electrochemical impedance spectroscopy (EIS) was conducted to further characterize interface stability of the BN/Na and Na electrode after 1 cycle, 100 and 500 cycles. As evidenced in Figure S5a (Supporting Information), EIS plots of the BN/Na electrode only show a mild increase of interfacial resistance upon cycling (≈100, 125, and 150 Ω after 1st cycle, 100 and 500 cycles respectively). In comparison, the pure Na electrode shows much larger resistance increases (≈125, 400, and 750 Ω after 1st cycle, 100 and 500 cycles respectively, Figure S5b, Supporting Information). The EIS studies again confirm the much better stability of the BN/Na electrode.

2.3. Morphology Evolution and Surface Analyses

Morphologies of the electrodes after cycling are shown in Figure 3. Upon cycling, the BN nanofiber structures are

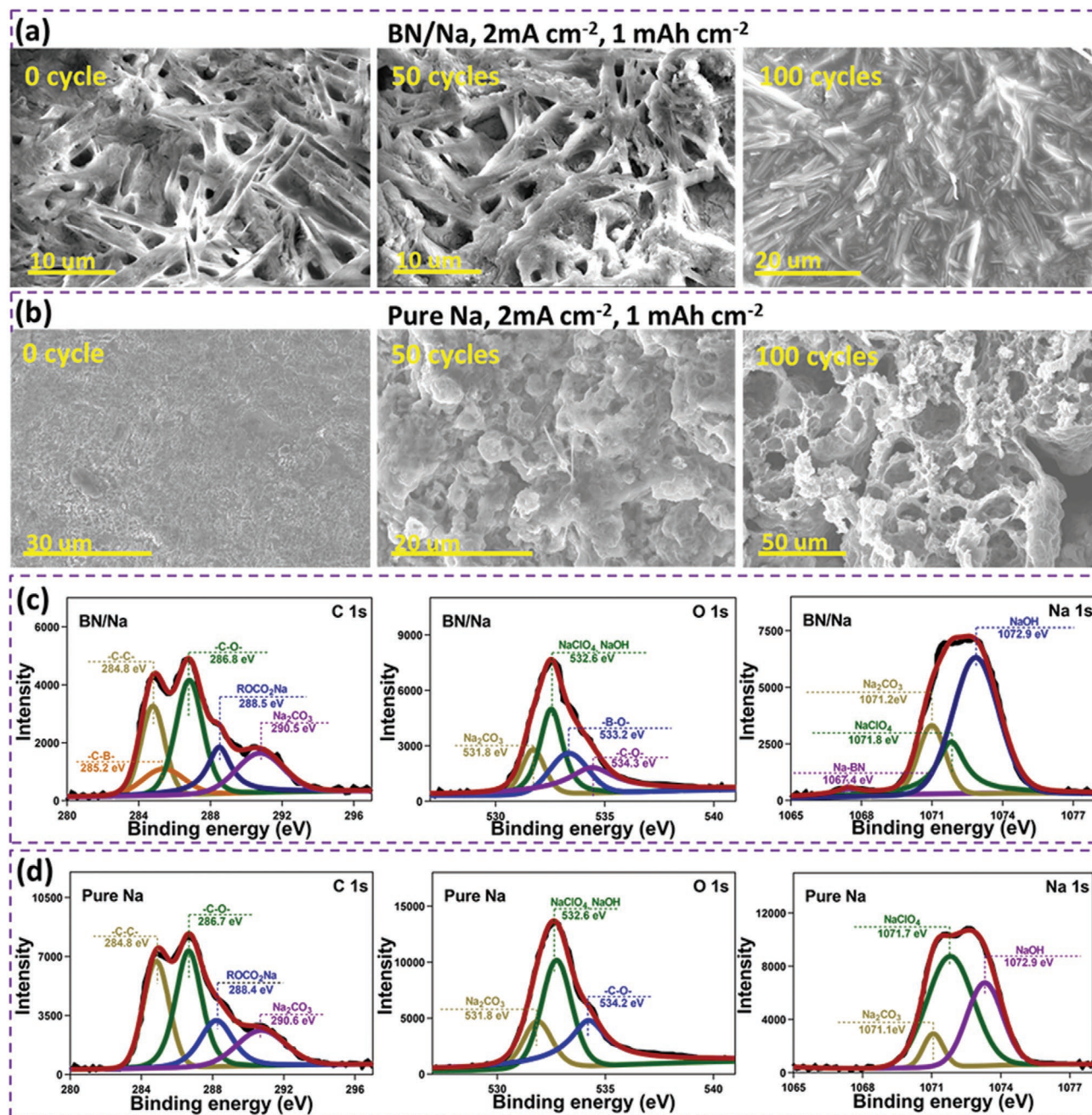


Figure 3. a) Morphology evolution characteristic of a pristine BN/Na electrode, a BN/Na electrode after 50 and 100 cycles at the current rate of 2 mA cm⁻². b) Morphology evolution images of a pure Na electrode after 0, 50, and 100 cycles at the current rate of 2 mA cm⁻². High-resolution XPS characterizations: c) C 1s, O 1s, and Na 1s of the BN/Na electrode after 100 cycles; d) C 1s, O 1s, and Na 1s of the pure Na electrode after 100 cycles.

basically maintained (Figure 3a). At the same time, roughness of the electrode surface decreases and the pores between the nanofibers are kind of filled upon cycling. We also used cross-section SEM to study the morphologies. As shown in Figure S6a (Supporting Information), BN fibers are mainly embedded in the top ≈10–20 μm of the sodium sheet. It can also be seen that upon cycling, original empty space between the BN fibers was mostly filled by sodium metal leading to a much smoother surface. On the other hand, the pure Na electrode shows

increasing roughness and porosity upon cycling (Figure 3b). As further verified by Figure S6b (Supporting Information), porous and dead Na of pure Na metal anode increased sharply from 0 to 100 cycles. These results are in good agreement with the electrochemical cycling stability in Figure 2. After cycling at higher current densities of 3 and 5 mA cm⁻² (Figure S7a,b, Supporting Information), the pure Na and the BN/Na electrodes undergo similar morphology changes as compared with these cells tested at 2 mA cm⁻² (Figure 3a,b). Coupled with the

electrochemical tests, this distinct morphology evolution characteristic further certifies that dendritic and rough structures are detrimental to Na-ion deposition properties, which not only deteriorate Coulombic efficiency, but also cause short circuit risks and damage the cells' structural integrity.^[19]

In parallel with morphology analysis, interface chemistry of the cycled BN/Na and Na electrodes were analyzed with XPS (Figure 3c,d and Figure S8, Supporting Information). In the C 1s spectrum of the BN/Na electrode (Figure 3c), peaks located at 284.8, 285.2, 286.8, 288.5, and 290.5 eV are ascribed to $-C-C-$, $-C-B-$, $-C-O-$, sodium organic complex ($ROCO_2Na$) and Na_2CO_3 respectively. The oxygen peaks at 531.8, 532.6, 533.2, and 534.3 eV corresponds to Na_2CO_3 , $NaOH/NaClO_4$, $-B-O-$ and $-C-O-$. For the Na 1s spectrum (Figure 3c), the typical peaks for Na_2CO_3 , $NaClO_4$ and $NaOH$ could be observed at 1071.2, 1071.8, and 1072.9 eV, respectively.^[9b] Intriguingly, an inconspicuous peak located at 1067.4 eV might result from interaction between Na and BN matrix. The cycled Na anodes do show similar peaks with

those in the cycled BN/Na anode except that there are no signal from B and N related peaks (Figure 3d). Other than this difference, signals from the decomposition products of the carbonate electrolyte are also much stronger in the spectra of the pure Na anode. This suggests that the electrolyte is less stable on the Na electrode.^[10a]

2.4. DFT and FES Simulation

To further understand the role of the BN nanofibers on Na-ion plating/stripping, density functional theory (DFT) and finite element simulation (FES) were performed to clarify the binding energy, charge differential density as well as smoothing effect. Charge differential density of the Na-ion on pure BN and defective BN are presented in Figure 4a–d. Clearly, no obvious interaction can be found between Na atom and a perfect BN lattice due to a low binding energy (-0.01 eV, Figure 4a). However, BN with N, B vacancies and BN divacancy significantly improves

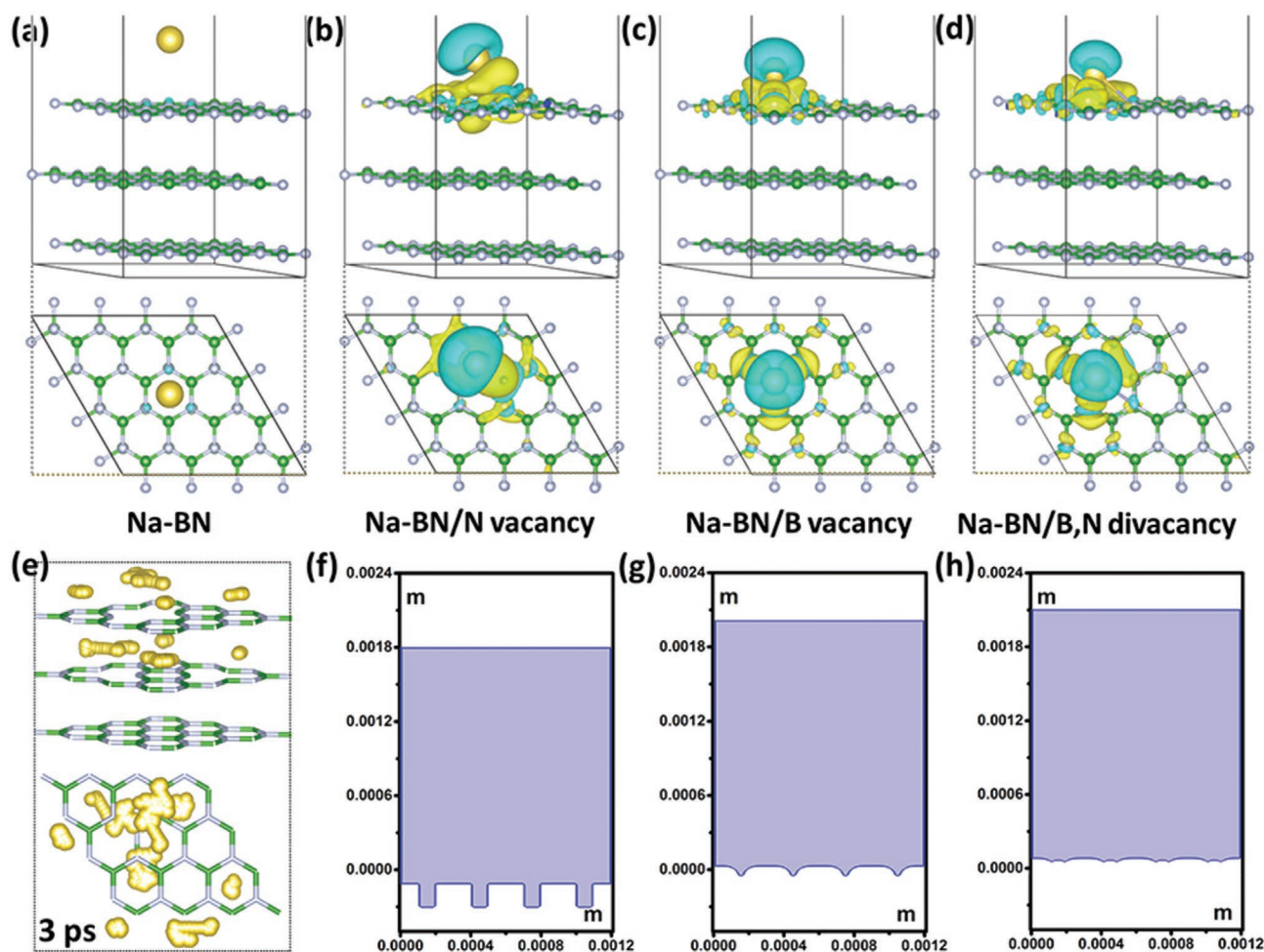


Figure 4. Charge differential density images of Na with a) pure BN, b) BN with N atom vacancy, c) BN with B vacancy and d) BN with B, N divacancy. e) Na-ion diffusion kinetics in BN lattice with B,N divacancy at 300 K for 3 ps. f–h) Comsol analysis of BN modified Na metal electrode surface morphology evolution characteristic at various reaction time (0, 2000, and 3600 s). White color represents Na metal, and blue color stands for electrolyte, x and y axis are electrode size.

the chemical interaction and enhances the charge transfer between Na atom and defective BN matrix (Figure 4b–d). The corresponding binding energies between Na atom and BN with N, B vacancies and BN divacancy are -0.41 , -0.36 , and -0.62 eV respectively. These enhanced interactions could facilitate Na-ion adsorption and electrolyte penetration.^[20] Besides the binding effect calculation, AIMD simulations were performed to assess interfacial Na-ion diffusion kinetics. Figure 4e and Figure S9 (Supporting Information) demonstrates possible Na-ion migration pathways ranging from 0 to 3 ps. The results show that even at a relative low temperature ($k_B T \approx 25$ meV), Na ions can easily hop from one energetic favorable site to another site near the B, N defects, suggesting a low Na-ion activation barrier and faster diffusion kinetics, and this would greatly conduce to smoothen Na-ion plating stripping process. To confirm this, we further simulate morphology evolution of a BN/Na and a Na electrode with an initial tooth-like protrusions using FES. For the BA/Na electrode, deposition of Na metal would gradually reduce the height of the teeth (Figure 4f–h) and the teeth nearly disappear at the end of the simulation. Adversely, Na ions prefer to deposit on the protruded teeth on the pure Na metal electrode. Deposition of Na metal would increase the height of the teeth and thus roughness of the electrode surface and finally destroy the cell (Figure S10, Supporting Information).^[21] To furthermore clarify this, current density distribution of the Na metal anode surface was also studied. It was clearly that current density distributed non-uniformly on the Na metal electrode surface and the current density is the highest at corners (Figure S11, Supporting Information). Subsequent electro-deposition will then further magnify the corners. In contrast, as shown in Figure 4f–h, BN layer tends to smooth the Na-ion deposition and the surface of the BN/Na is getting smoother upon repeated cycling. Hence, current density distribution is becoming uniform for Na-ion deposition, leading to a stable Na-ion deposition behavior. All these calculation results clearly reveal the regulation mechanism of BN on Na-ion plating/stripping process, matching well with prior chemical and electrochemical analysis.

2.5. Sodium-Metal Full Batteries

Full cells pairing the BN/Na anode with a good cathode need to be constructed for further evaluation. $\text{Na}_3\text{V}_2\text{O}_2(\text{PO}_4)_2\text{F}$, by virtue of its high theoretical specific capacity (130 mAh g^{-1}), attractive voltage platforms (3.6 and 4.0 V vs sodium) and robust 2D open Na-ion transport channel, is a promising candidate for SIBs cathode. Figure 5a illustrates crystal structure of $\text{Na}_3\text{V}_2\text{O}_2(\text{PO}_4)_2\text{F}$. Particularly, two irregular $[\text{VO}_3\text{F}]$ octahedrons connected by corner-sharing F atom form a $[\text{V}_2\text{O}_{10}\text{F}]$ bi-octahedron structural unit and these units are bridged by $[\text{PO}_4]$ tetrahedron, thus expanding into layered structure, with Na^+ ions accommodated between layers. Nevertheless, unsatisfactory electronic conductivity (bandgap ≈ 2.1 eV) as shown in Figure 5b still imperils its large-scale application. This problem was addressed by forming a composite with Mxene ($\text{Ti}_3\text{C}_2\text{X}$), which has a high electric conductivity endowed from its ultrasmall bandgap.^[22] Hexagonal Ti_3AlC_2 powder, which owns a close-packed layer structure, was utilized as the raw material

(Figure S12a,b, Supporting Information). $\text{Ti}_3\text{C}_2\text{X}$ nanosheets, as presented in Figure 5c and Figure S13a,b (Supporting Information), were fabricated via a facile HF-etching method. It is worth noting that typical (002) peak shifts from 9.51° (Figure S12a, Supporting Information) to 5.41° (Figure S13a, Supporting Information), indicating the formation of well-stripped $\text{Ti}_3\text{C}_2\text{X}$ nanosheets.^[23] Afterward, NVPOF/M composite was synthesized via a hydrothermal reaction. Figure 5d shows XRD pattern of the as-synthesized NVPOF/M composite. The peak located at 5.36° is ascribed to the (002) lattice plane of the $\text{Ti}_3\text{C}_2\text{X}$ nanosheets, while all the other peaks correspond to tetragonal $\text{Na}_3\text{V}_2\text{O}_2(\text{PO}_4)_2\text{F}$ with a space group of $I4/mmm$ ($a = b = 6.37028 \text{ \AA}$, $c = 10.6365 \text{ \AA}$, $\alpha = \beta = \gamma = 90^\circ$). Energy dispersive X-ray spectroscopy (EDS) shows the strong Na, V, O, P, F, Ti, and C element signal (Figure 5e), coupled with elemental mapping images (Figure 5f) and TEM images (Figure S14a–c, Supporting Information), reaffirming that $\text{Na}_3\text{V}_2\text{O}_2(\text{PO}_4)_2\text{F}$ nanoparticle is attached uniformly among the $\text{Ti}_3\text{C}_2\text{X}$ substrates and this is favorable for redox kinetics and electrochemical utilization.

We next investigate the dynamic electrochemistry of the full cells. It is observed that these cells can deliver a high charge/discharge capacity of $128/124 \text{ mAh g}^{-1}$ (Figure 6a), corresponding to 1.9 electron transfer and result in superior initial Coulombic efficiency of 96%. Such high initial materials utilization is mainly attributed to the novel cells' configuration and improved electron transfer. Figure 6b demonstrates rate capability of the full cell, which yield stable charge-storage capacities of 125, 111, 99, 81, and 61 mA h g^{-1} at 0.05, 0.1, 1, 2, and 5 C respectively. After cycling at 5 C, the cells still retain an acceptable capacity of 108 mA h g^{-1} at 0.05 C rate, suggesting the excellent rate capability of this full cell. When cycling at 0.05 and 5 C, the full cells exhibited long-term cycling stability with high capacity retention of 81% and 78% respectively after 500 cycles (Figure 6c,d), corresponding to slow decay of 0.038% and 0.044% per cycle. Meanwhile, almost unchanged cyclic voltammogram (CV) curves from 1st to 200th cycles at a scan rate of 1 mV s^{-1} (Figure S15, Supporting Information) further indicate the high rate cycling stability of the full cell. Combined with high output voltage (inset Figure 6a), these novel NVPOF/M-BN/Na full cell gives rise to a promising energy density of $\approx 420/240 \text{ Wh kg}^{-1}$ at 0.05/5 C (based on cathode).

A particular comparison analysis of the full cell with pure Na metal anode is summarized in Figure S16 (Supporting Information). As shown in Figure S16a (Supporting Information), these control cell demonstrates a relatively unstable CV profiles from 1st to 200th cycles at a high scan rate of 1 mV s^{-1} . Albeit similar rate capabilities of 123, 108, 96, 82, and 60 mA h g^{-1} at 0.05, 0.1, 1, 2, and 5 C respectively are found in Figure S16b (Supporting Information), large fluctuations of coulombic efficiency are detected especially at high rate 5 C, thus largely explained by the unstable Na-ion deposition on NMA at high current density. Cycled at 5 C (Figure S16c, Supporting Information), these cells experience a similar cycling and efficiency instability, again testifying above statement. The present study demonstrates that BN/Na anode has excellent potential for the building of advanced sodium metal batteries.

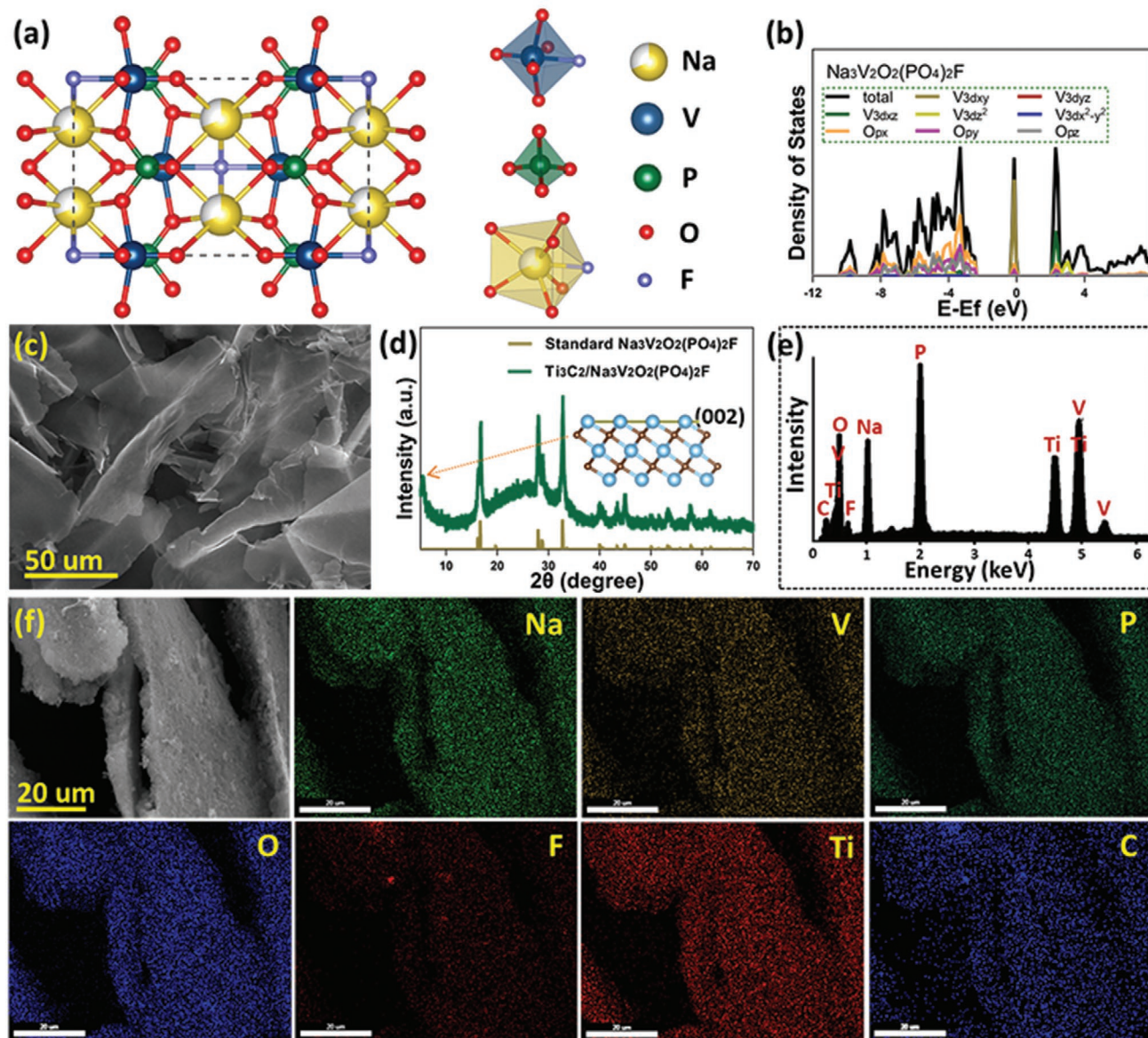


Figure 5. a) Crystal structure of Na₃V₂O₂(PO₄)₂F. b) Density of states of Na₃V₂O₂(PO₄)₂F. c) SEM image of Ti₃C₂X (X = -OH or -F). d, e) XRD and EDS profiles of the as-obtained Na₃V₂O₂(PO₄)₂F/Ti₃C₂X composite. f) Elemental mapping characterization of the as-synthesized Na₃V₂O₂(PO₄)₂F/Ti₃C₂X composite.

3. Conclusion

In conclusion, a porous BN nanofibers modified sodium metal (BN/Na) electrode is fabricated via a simple spin-coating method. Upon cycling, a symmetrical BN/Na cell achieves a superior rate capability and competitive long-cycling life over 600 h at 10 mA cm⁻² with carbonate-based electrolyte. DFT and FES simulations demonstrate that the presence of defective BN not only facilitates Na-ion adsorption and diffusion, but also prompts smoothing upon extensive Na-ion plating/stripping cycles. Moreover, a full cell coupled BN/Na anode with NVPOF/M cathode offers a highly stable capacity of 125/63 mA h g⁻¹ (≈420/240 Wh kg⁻¹) at 0.05/5 C rate over 500 cycles, thus

making BN/Na anode very applicable for the establishment of high-energy-density sodium metal full batteries.

4. Experimental Section

Synthesis of BN Nanofibers: Typically, 25 mmol boric acid and 10 mmol melamine were added into 100 mL deionized water simultaneously, which were stirred for 1 h to achieve a uniform mixture. Afterward, the solution was gradually heated to 80 °C and maintained at that temperature for 1 h to obtain a transparent solution. Next, a gel-like BN precursor was acquired (boric acid and melamine molecules are connected by hydrogen bond as shown in Figure 1a) by cooling the solution naturally to room temperature. After a freeze-drying

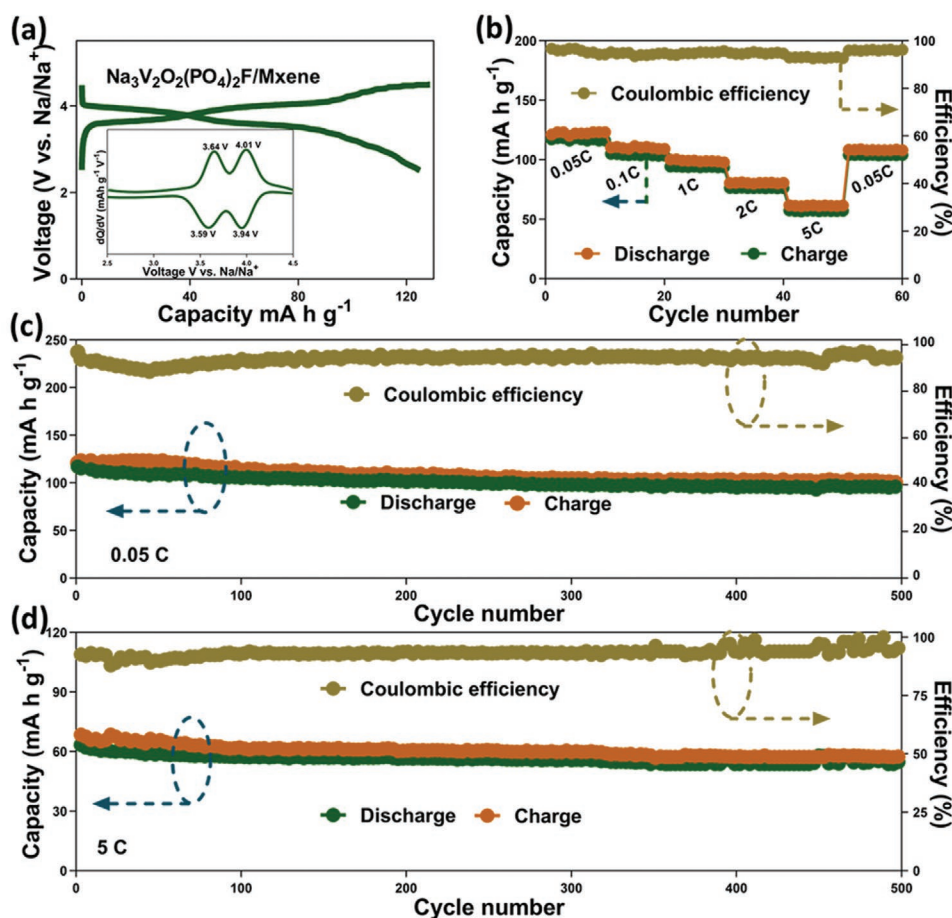


Figure 6. Electrochemical performance of the full cell paired $\text{Na}_3\text{V}_2\text{O}_2(\text{PO}_4)_2\text{F}/\text{Ti}_3\text{C}_2\text{X}$ cathode with BN/Na anode: a) voltage profile of the full cell cycled at 0.05 C within 2.5–4.5 V (inset is dQ/dV curve); b) rate capability at 2.5–4.5 V; c,d) long-cycling performance at 0.05 and 5 C respectively within 2.5–4.5 V.

step, BN nanofibers were finally synthesized through dehydration condensation and annealing the precursor at 1000 °C for 5 h under argon atmosphere.

Synthesis of BN Nanofibers Modified Na Metal: Firstly, 50 mg as-synthesized BN nanofibers were dispersed into 5 mL 1 M NaClO_4 EC/PC electrolyte (inset Figure S1d, Supporting Information). Subsequently, the solution was coated on a Na metal surface via spin-coating method with 200 rpm min^{-1} for 20 min in a glove box. The obtained BN/Na was dried naturally in the glove box and then pressed into the surface of Na circular sheets (1 cm^2) using a steel rolling rod (Figure S2, Supporting Information). Average weights of Na sheets and BN/Na sheets are respectively 11.68 and 12.71 mg (inset Figure S3, Supporting Information).

Fabrication of Mxene ($\text{Ti}_3\text{C}_2\text{X}$) Nanosheets: In detail, 1 g Ti_3AlC_2 and 1 g LiF powders were slowly added into 10 mL hydrochloric acid for more than 10 min to prevent thermal accumulation in a PTFE container. Solution was then continuously stirred in 60 °C for 24 h to remove Al atom layer. The obtained mixture was then centrifuged at 3500 rpm for 10 min to collect deposits. Next, the deposits were thoroughly cleaned with deionized water until reaching a neutral pH. Finally, the solid products were dispersed into ultrapure water under ultrasonic condition at least 4 h to obtain the dark green Mxene nanosheets dispersion solution ($\approx 1 \text{ mg mL}^{-1}$).

Fabrication of $\text{Na}_3\text{V}_2\text{O}_2(\text{PO}_4)_2\text{F}/\text{Mxene}$ ($\text{Ti}_3\text{C}_2\text{X}$) Composite: Specially, 5 mL of the above Mxene solution and 0.1 g CTAB were gradually added into 45 mL DMF and sonicated for 30 min. A solution containing

0.001 mol NH_4VO_3 , 0.0015 mol NaF, and 10 mL deionized water was then drop-wisely added into above solution. Subsequently, 10 mL $(\text{NH}_4)_2\text{HPO}_4$ (0.001 mol) aqueous solution was slowly added into above solution to ensure the chemical molar ratio Na: V: P = 3: 2: 2: 3. This mixture was finally transferred into 60 mL Teflon autoclaves and treated at 180 °C for 12 h to obtain the final product.

Materials Characterization: Detailed microstructure and elemental analysis were characterized with a QUANTRO S scanning electron microscope. HRTEM images and corresponding electron diffraction tests were performed on a JEOL-2010F TEM instrument. XRD patterns were determined on a X-ray diffractometer with $\text{Cu-K}\alpha 1$ radiation ($\lambda = 1.5405 \text{ \AA}$). X-ray photoelectron spectroscopy (XPS, VG ESCALAB 220i-XL) was utilized to analyze surface and bonding characteristics of BN and cycled sodium electrode. All the cycled electrodes were sealed in argon-filled devices for measurements to avoid air oxidation.

Electrochemistry Measurements: Voltage profiles and long-term cycling performances of Na||Na, BN/Na||BN/Na symmetric cells and $\text{Na}_3\text{V}_2\text{O}_2(\text{PO}_4)_2\text{F}/\text{Ti}_3\text{C}_2\text{X}$ ||Na or BN/Na full metal cells employing 1 M NaClO_4 EC/PC electrolyte and glass fiber separator were tested with a MACCOR battery cyler. Sodium-ion plating/stripping performances of Na||Na, BN/Na||BN/Na symmetric cells was evaluated from 2 to 10 mA cm^{-2} at 1 mA h cm^{-2} . Rate and cycling capability of $\text{Na}_3\text{V}_2\text{O}_2(\text{PO}_4)_2\text{F}/\text{Ti}_3\text{C}_2\text{X}$ ||Na or BN/Na full metal cells were examined from 0.05 to 5 C rates over 500 cycles. Electrochemical impedance spectroscopy (EIS) measurements were done with an impedance analyzer (Zennium pro) in a frequency ranging from 100 mHz to 1 MHz.

DFT Calculation and COMSOL Simulation: Several supercells (3 layers, 97 atoms in total as shown in Figure 4a) containing N, B vacancy (Figure 4b,c, 96 atoms in total) and B/N-divacancy (Figure 4d, 95 atoms in total) were fabricated and the vacuum layer is 15 Å. Na-ion adsorption behavior and charge differential density were analyzed according to density functional theory (DFT) by utilizing Vienna ab Initio Simulation Package (VASP) with projector augmented plane-wave method.^[24] Exchange-correlation functional of Perdew–Burke–Ernzerhof was employed to deal with the exchange–correlation interaction between itinerant electrons. Relative weak van der Waals interactions between Na-ion and BN matrix were treated by semi-empirical London dispersion corrections of Grimme^[25] Cut-off energy of 600 eV was utilized for all the simulations to ensure energy and structure convergence. Monkhorst–Pack k-point mesh was $6 \times 6 \times 1$. For the electronic structure calculation of $\text{Na}_3\text{V}_2\text{O}_2(\text{PO}_4)_2\text{F}$, a U value of 4.2 eV needs to be adopted to deal with localized 3d electrons of V element and the Monkhorst–Pack k-point mesh is $3 \times 3 \times 3$. Ab-initial molecular dynamics (AIMD) calculations were then conducted to illustrate the Na-ion diffusion kinetics in defective BN lattice (3 layers, 8 Na atoms and 92 B,N atoms in total as shown in Figure 4e). This simulation was implemented by Gaussians and Plane Wave method (GPW) embedded in the CP2K software package at single Γ point with an energy cutoff of 400 Ry.^[26] The simulation is lasted for 3 ps with a time step of 1 fs, and the equilibrium temperature is 300 K controlled by Nose–Hoover thermostat. COMSOL simulation was utilized to demonstrate smoothing effect of BN on Na-ion plating/stripping process. For the Na-ion electro-deposition process, the Butler–Volmer equation for Na-ion deposition reaction is displayed as following^[21,27]

$$i = -c \times \exp(-\alpha F\eta/RT) i_0 \quad (1)$$

However, the B–V equation should consider surface coverage effect with the presence of BN, which is empirically expressed by

$$i = -(c + a\theta) \times \exp(-\alpha + b\theta)F\eta/RT) i_0 \quad (2)$$

Na metal surface electro-deposition rate is given according to Faraday's law of electrolysis

$$V = -Mi/\rho F n \quad (3)$$

In these formulas, i is the Na-ion deposition current density, i_0 stands for the exchange current density. c represents the Na-ion concentration in electrolyte, α is the corresponding electron transfer coefficient. η is over-potential versus standard Na^+/Na . F and R are the faraday constant and molar gas constant respectively. a (coefficient of BN surface coverage), and b (improved electronic transfer coefficient of the BN). M and ρ are molecular weight and the density of Na metal, while n is the transferred electrons.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Keywords

boron nitride nanofibers, density functional theory simulations, FES analysis, $\text{Na}_3\text{V}_2\text{O}_2(\text{PO}_4)_2\text{F}/\text{Ti}_3\text{C}_2\text{X}$ cathode, sodium metal anodes

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